

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet

range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

ANSWERS

➤ Question 1

Problem

An organization has been allocated the IP address **192.168.10.0/24**. It needs four subnets for the following departments:

- HR – 50 hosts
- Sales – 25 hosts
- IT – 10 hosts
- Admin – 5 hosts

Determine:

1. Suitable subnet mask
2. Network Address
3. Valid Host Range
4. Broadcast Address
5. Remaining unused IP addresses

Solution

Step 1: Calculate Required Subnets

Department Hosts Required Prefix Usable Hosts

HR	50	/26	62
Sales	25	/27	30
IT	10	/28	14
Admin	5	/29	6

HR Department (/26)

- Network Address: **192.168.10.0**
- Host Range: **192.168.10.1 – 192.168.10.62**
- Broadcast Address: **192.168.10.63**
- Usable Hosts: **62**

Sales Department (/27)

- Network Address: **192.168.10.64**
- Host Range: **192.168.10.65 – 192.168.10.94**
- Broadcast Address: **192.168.10.95**
- Usable Hosts: **30**

IT Department (/28)

- Network Address: **192.168.10.96**
- Host Range: **192.168.10.97 – 192.168.10.110**
- Broadcast Address: **192.168.10.111**
- Usable Hosts: **14**

Admin Department (/29)

- Network Address: **192.168.10.112**
- Host Range: **192.168.10.113 – 192.168.10.118**
- Broadcast Address: **192.168.10.119**
- Usable Hosts: **6**

Remaining Unused IP Address Range

192.168.10.120 – 192.168.10.255

Final Answer

Department	Network Address	Host Range	Broadcast Address	Usable Hosts
HR	192.168.10.0	192.168.10.1 – 192.168.10.62	192.168.10.63	62
Sales	192.168.10.64	192.168.10.65 – 192.168.10.94	192.168.10.95	30
IT	192.168.10.96	192.168.10.97 – 192.168.10.110	192.168.10.111	14
Admin	192.168.10.112	192.168.10.113 – 192.168.10.118	192.168.10.119	6

Unused IP Range: 192.168.10.120 – 192.168.10.255

➤ Question 2

Problem

A company has the network **10.10.0.0/16**. Allocate subnets using VLSM for:

- Engineering – 500 hosts
- Marketing – 200 hosts
- Finance – 100 hosts
- Support – 50 hosts

Find:

1. Network Address
2. Host Range
3. Broadcast Address
4. Usable Hosts

Solution

Department Hosts Required Prefix Usable Hosts

Engineering	500	/23	510
Marketing	200	/24	254
Finance	100	/25	126
Support	50	/26	62

Engineering (/23)

- Network Address: **10.10.0.0**
- Host Range: **10.10.0.1 – 10.10.1.254**
- Broadcast Address: **10.10.1.255**
- Usable Hosts: **510**

Marketing (/24)

- Network Address: **10.10.2.0**
- Host Range: **10.10.2.1 – 10.10.2.254**
- Broadcast Address: **10.10.2.255**
- Usable Hosts: **254**

Finance (/25)

- Network Address: **10.10.3.0**
- Host Range: **10.10.3.1 – 10.10.3.126**
- Broadcast Address: **10.10.3.127**
- Usable Hosts: **126**

Support (/26)

- Network Address: **10.10.3.128**
- Host Range: **10.10.3.129 – 10.10.3.190**
- Broadcast Address: **10.10.3.191**
- Usable Hosts: **62**

Final Answer

Department	Network Address	Host Range	Broadcast Address	Usable Hosts
Engineering	10.10.0.0	10.10.0.1 – 10.10.1.254	10.10.1.255	510
Marketing	10.10.2.0	10.10.2.1 – 10.10.2.254	10.10.2.255	254
Finance	10.10.3.0	10.10.3.1 – 10.10.3.126	10.10.3.127	126
Support	10.10.3.128	10.10.3.129 – 10.10.3.190	10.10.3.191	62

➤ Question 3

Problem

An organization has been allocated the IP block **172.16.0.0/16**. The departments require the following subnets:

- HR → /25
- Sales → /26
- IT → /27
- Admin → /28

Determine:

1. Network Address
2. Valid Host IP Range
3. Broadcast Address
4. Number of Usable Hosts
5. Remaining Unused IP Address Range

Formula

1. Block Size

$$\text{Block Size} = 256 - \text{Last Octet of Subnet Mask}$$

2. Usable Hosts

$$\text{Usable Hosts} = 2^{(32 - \text{Prefix})} - 2$$

3. Broadcast Address

$$\text{Broadcast Address} = \text{Next Network Address} - 1$$

Calculations

HR Department (/25)

Subnet Mask = 255.255.255.128

Block Size

$$256 - 128 = 128$$

Network Address = 172.16.0.0

Broadcast Address = 172.16.0.127

Host Range = 172.16.0.1 – 172.16.0.126

Usable Hosts

$$2^7 - 2 = 128 - 2 = 126$$

Sales Department (/26)

Subnet Mask = 255.255.255.192

Block Size

$$256 - 192 = 64$$

Network Address = 172.16.0.128

Broadcast Address = 172.16.0.191

Host Range = 172.16.0.129 – 172.16.0.190

Usable Hosts

$$2^6 - 2 = 64 - 2 = 62$$

IT Department (/27)

Subnet Mask = 255.255.255.224

Block Size

$$256 - 224 = 32$$

Network Address = 172.16.0.192

Broadcast Address = 172.16.0.223

Host Range = 172.16.0.193 – 172.16.0.222

Usable Hosts

$$2^5 - 2 = 32 - 2 = 30$$

Admin Department (/28)

Subnet Mask = 255.255.255.240

Block Size

$$256 - 240 = 16$$

Network Address = 172.16.0.224

Broadcast Address = 172.16.0.239

Host Range = 172.16.0.225 – 172.16.0.238

Usable Hosts

$$2^4 - 2 = 16 - 2 = 14$$

Remaining Unused IP Range

Start = 172.16.0.240

End = 172.16.255.255

Final Answer

Department	Network Address	Host Range	Broadcast Address	Usable Hosts
HR	172.16.0.0	172.16.0.1 – 172.16.0.126	172.16.0.127	126
Sales	172.16.0.128	172.16.0.129 – 172.16.0.190	172.16.0.191	62
IT	172.16.0.192	172.16.0.193 – 172.16.0.222	172.16.0.223	30
Admin	172.16.0.224	172.16.0.225 – 172.16.0.238	172.16.0.239	14

Remaining Unused IP Address Range:**172.16.0.240 – 172.16.255.255****Interpretation**

The 172.16.0.0/16 network is subnetted efficiently using VLSM. Each department receives an appropriate subnet size while leaving a large address space available for future expansion.

➤ Question 4**Problem**

A company uses the IP address 172.16.10.0/24 with a /27 subnet mask.

Find:

1. Total number of subnets
2. Number of usable hosts per subnet
3. Subnet for IP 172.16.10.78
4. Network Address
5. Broadcast Address
6. Check whether 172.16.10.95 belongs to the same subnet

Formula**Number of Subnets**

$$\text{Number of Subnets} = 2^{\text{Borrowed Bits}}$$

Borrowed Bits

$$27 - 24 = 3$$

Usable Hosts

$$2^{(32-27)} - 2$$

Block Size

$$256 - 224 = 32$$

Calculations**Step 1: Number of Subnets**

$$2^3 = 8$$

Total Subnets = 8**Step 2: Usable Hosts**

$$2^5 - 2$$

$$32 - 2 = 30$$

Usable Hosts = 30**Step 3: Subnet Table**

Network Address	Host Range	Broadcast
172.16.10.0	172.16.10.1–30	172.16.10.31
172.16.10.32	172.16.10.33–62	172.16.10.63
172.16.10.64	172.16.10.65–94	172.16.10.95
172.16.10.96	172.16.10.97–126	172.16.10.127
172.16.10.128	172.16.10.129–158	172.16.10.159

Network Address	Host Range	Broadcast
172.16.10.160	172.16.10.161–190	172.16.10.191
172.16.10.192	172.16.10.193–222	172.16.10.223
172.16.10.224	172.16.10.225–254	172.16.10.255

Step 4: Find the Subnet for 172.16.10.78

The IP **172.16.10.78** falls in the subnet:

- **Network Address:** 172.16.10.64
- **Host Range:** 172.16.10.65 – 172.16.10.94
- **Broadcast Address:** 172.16.10.95

Step 5: Verify 172.16.10.95

- **172.16.10.95** belongs to the same subnet.
- It is the **broadcast address**, so it **cannot be assigned to a host**.

Final Answer

Parameter	Answer
Total Subnets	8
Usable Hosts per Subnet	30
Network Address	172.16.10.64
Host Range	172.16.10.65 – 172.16.10.94
Broadcast Address	172.16.10.95
172.16.10.95 Status	Broadcast Address (same subnet, not a valid host)

Interpretation

Using a /27 subnet mask divides the **172.16.10.0/24** network into **8 equal subnets**, each supporting **30 usable hosts**. The IP address **172.16.10.78** is a valid host in the **172.16.10.64/27** subnet, while **172.16.10.95** is the broadcast address reserved for that subnet.

➤ Question 5

Problem

A company has the network **192.168.50.0/24** and uses a /26 subnet mask.

Find:

1. Total number of subnets
2. Number of usable hosts per subnet
3. Identify the subnet for **192.168.50.130**
4. Network Address
5. Broadcast Address
6. Check whether **192.168.50.190** belongs to the same subnet

Solution

Step 1: Number of Subnets

Borrowed bits:

$$26 - 24 = 2$$

$$2^2 = 4$$

Total Subnets = 4

Step 2: Usable Hosts

$$2^{(32-26)} - 2 = 64 - 2 = 62$$

Usable Hosts = 62

Step 3: Subnet Table

Network Address	Host Range	Broadcast Address
192.168.50.0	192.168.50.1 – 62	192.168.50.63
192.168.50.64	192.168.50.65 – 126	192.168.50.127
192.168.50.128	192.168.50.129 – 190	192.168.50.191
192.168.50.192	192.168.50.193 – 254	192.168.50.255

Step 4: IP Address Analysis

IP **192.168.50.130** belongs to:

- **Network Address:** 192.168.50.128
- **Host Range:** 192.168.50.129 – 192.168.50.190
- **Broadcast Address:** 192.168.50.191

IP **192.168.50.190** is a **valid host** in the **same subnet**.

Final Answer

Parameter	Answer
Total Subnets	4
Usable Hosts per Subnet	62
Network Address	192.168.50.128
Host Range	192.168.50.129 – 192.168.50.190
Broadcast Address	192.168.50.191
192.168.50.190 Status	Valid host in the same subnet